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Multiple regression Analysis

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Conceptual framework

The European Social Survey is a cooperative effort that breaks more than one limit in quantitative statistics. Most obviously, it standardizes across languages and borders, a practice only possible by a common understanding of numerical data. Then, it pulls from vastly singular lives to ordinate general truths. But what I found most shocking was the bottom-line process through which this information is gathered, more specifically the ordering of questions. Ensuring all participants are comfortable enough to answer intrusive questions requires delicate wording and minute ordering of questions that flow logically and progressively get to sensible points that would not get disclosed from the get-go.

I praise this questionnaire for I too was so soothed by it that I struggled to take it apart. It is hard to motivate a research design when all the data is thematically linked from the start, causing biased answers from respondents trying to be consistent with their previous answers. Nevertheless, forcing myself to look at overarching ideas proved useful: I isolated variables that did not refer to anything physical, but instead dealt with spiritual identity. Among these, I was particularly interested in establishing a respondent's proximity with a set of values. Is success important? Does one act to be praised? If it is the case, can we consider this external recognition a main driver of our world? To consider these questions I chose to examine their impact on different fringes of Danish society. The ethnic background related gap I chose to measure is also related to spiritual identity: religion. What good brings superficial demonstrations of wealth and fancy job titles in the face of meaningful spiritual connection and enlightenment?

If answering any of the above questions is not the goal of this paper, finding out people's answers to them and why they think that way is precisely what I will try to do. Although figuring out the distribution of success driven people relatively to their religious beliefs won't be nearly as challenging as getting a grasp of what factors explain their position on the matter.

Consider the following hypothesis: non-religious people aspire more to be successful than religious people. As a note, throughout this essay “irreligious” “non-religious” and “religionless” will be used interchangeably to refer to all those not self-identifying as belonging to a religion regardless of their agnostic or atheist beliefs. I identify three possible explanations for the hypothesized attitude difference. First, a difference in esteem towards privileges could indicate that religious peoples do not strive for success as they have already found comfort in inherited social status (Bloechl 2021). Secondly, I postulate that the desire to be successful comes from an alternative form of self-improvement. Here, career-oriented aspirations would be more spiritually meaningful to the non-religious in face of the many delusions the modern world offers (going along declining rates of believer across all religions in recent decades). Lastly, I picture the internet and its media as a constant reminder of the success of others and as aggravating our desire for recognition, to which religious people would be less exposed (Moberg 2020).

Hence, here is the research question: is there a gap between religious and non-religious attitude towards success, and if so, can this gap be explained by differences in privileges, aspirations, and internet exposure? The importance of success is the dependent variable, religious status is the central gap, and content with privilege, active careerism and internet exposure are the independent variables.

Methodological design

The methodological design to investigate the research question is quite simple. I aim at comparing results from multiple regressions that include and exclude a range of variables to locate parameters having a bigger influence on the relationship between the central gap and the dependent variable. Proceeding as such assumes a certain plausibility that the hypotheses is true and provides no failsafe should the null hypothesis not be rejected. However, I count on the large sample size and craftiness of the necessary adjustments to keep the p-values within the determined threshold. Should p-values consistently rule out the compatibility of the findings with the hypothesis, turning to randomized controlled trials could revitalize the measure of random variability in the sample.

For this, I take variables from the ESS9 data set substantively similar to the concepts I want to test for. “rlgblg” asks if the respondent belongs to a religion, I recoded it as “breligion”: a binary dummy variable excluding those refusing to answer or not knowing (only 5 respondents have done this). “ipsuces” measures how strongly the respondent agrees that being successful is important and that they hope others will recognize their achievements. I recoded it as “success” but kept the original distribution of values on a scale of 1 to 6, where 1 is “Very much like me”, and 6 “Not like me at all”. For the independent variables, “sofrprv” asks if the respondent thinks those from high status families should enjoy privileges, which translates well the idea of path-dependency that I was seeking in the design (Bloechl 2021). Indeed, this controversial claim can be interpreted by religious people as gratefulness for what we own and by non-religious as unfair and underserved success (recoded as “privilege”). For the internet consumption, “netustm” tracks how many minutes per day one spends on the internet (recoded as “internet_activity”), and for active careerism “atncrse” (recoded as “careerism”) asks if participants have taken any course or attended any lecture or conference to improve your knowledge or skills for work. Improving knowledge could be deemed too general, but since the question addresses whether these activities have been

done for work specifically, it is a step towards climbing the corporate ladder that exemplifies well the idea in the design.

When recoding, I attempted to rescale the independent variables since their format is so different. Privilege is measured on a scale of 1 to 5 with 1 as those agreeing strongly that inherited privileges are deserved and 5 being those disagreeing strongly with that claim. Internet activity is measured in minutes, and careerism is a yes or no question. Finding a common ground for these 3 different types of variables is challenging, and I resorted to using z-standardization and to finding interaction terms between continuous (internet activity) categorical (privilege) and binary (careerism) variables.

In terms of causal assumptions, this analysis assumes that religiosity is exogenous to the attitude towards success. Furthermore, the included covariables should cover all cofounders jointly affecting religiosity and the importance of success within the limits of this design. Although there could be socio-cultural factors at play in this relationship, the goal being to pin down values and aspirations motivating an individual's alignment with the dependent variable given their religious status, the identified covariables should suffice to reach a conclusion.

Though there is still the possibility of a reverse causality in religiosity, the historical context of Denmark, its church-state separation and stable religious trends makes this claim less likely than in other countries. Besides, robustness of the results given the large sample size of the European Social Survey and the thorough methodology with which the survey was conducted should minimize post treatment bias. Still, in future designs, considering religiosity as a more complex phenomenon than a binary value and including unobserved cofounders such as personality traits or family influences could produce less biased results.

Since this design is purely observational and only draws from existing data, many parameters leave room for improvement. One of the major issues with the data set surrounds unobserved differences. Indeed, it is important not to mistake correlation for causation and make naive comparisons in the face of selection bias. A concrete example of that in this design would be respondents self-selecting themselves into the religious categories depending on the context in which the question is asked. There could be religious relatives present during the respondent's answer, they could feel compelled to answer a certain way due to the interviewer's approach, in light of past responses given, or as a habit, or there could be a higher chance a religious respondent agrees to take the survey due to religion cultivating their charitable traits. Ideally, collecting specific data for this design would allow for appropriate measures ensuring minimal errors for this research. This however is not the only issue with pre-existing data sets: not getting to ask the questions means there is a possibility that unknown confounding variables make the correlation between the independent and the dependent variable spurious. Let's consider a variable measuring the degree to which the respondent was autonomous during their childhood. A high value would mean abandonment of the individual and their survival depending entirely on themselves, while a low value would depict a sheltered upbringing. While there is no similar question in the European Social Survey, this variable would directly interfere with both religious identity, as it is often inherited, and drive towards success and recognition, if fueled by traumatic fear of abandonment. Therefore, an ideal design would test the hypotheses while holding every other variable constant, an operation hardly - if ever - achievable in social sciences. Nevertheless, being aware of these confounders and mediators is already a great step in isolating the relation between the predictor and the outcome.

Thinking a step further, an ideal experimental design would allow researchers to manipulate religiosity in randomized control trials where participants are assigned values and react to the exposure to new beliefs. This would allow the research to not only identify the most impactful cofounders, but also to observe those that ultimately change the participants' opinion. For obvious ethical reasons, this is not a design that can be implemented.

Empirical analysis

The first OLS regression only includes one independent variable: the binary dummy variable that makes up the central gap of religiousness. Looking at the results, the intercept is 3.239: the reference group (non-religious respondents) have an average success score of 3.2, indicating moderate agreement that success is important. The religion variable has a p-value of 0.225 making it not statistically significant at the 0.05 significance level, and the R-squared score of 0.1% means religious status alone explains almost none of the variance in attitude towards success. This first regression therefore does not allow for the null hypothesis to be rejected. Statistical insignificance demands carefully interpreting the coefficients: the negative coefficient direction (-0.089) suggests that religious individuals care slightly less about success. All of this indicates that the factors not included in this model may better explain any potential gap.

Next, the second OLS regression includes all 3 independent covariables. A direct improvement from the first model is visible with r-squared measures: 7.4% of variations in attitudes towards success are explained by this regression. Also, with a P-value of 0.104, the results are still not statistically significant at the 0.005 level but show a clear improvement from the first model. Religion is on the right path to become a predictor of attitude towards success when

accounting for covariables more carefully. P-values for all other variables show statistical significance, showing they affect success. These covariables vary largely in coefficient: higher career aspirations lead to more enthusiasm towards success (coefficient of 0.348) but internet usage has a relatively null negative coefficient (-0.001), meaning more time online makes an individual even so slightly less likely to aspire for success.

Thirdly, I run another regression with all the variables, this time standardizing to allow for easier interpretation and to compare the results with the second regression (see Table 1). Standardizing the variables confirmed that careerism and internet activity—not religiosity—are the primary drivers of attitudes toward success. While the religious gap remained statistically insignificant (-0.086 SD, $p = 0.104$), careerism exhibited the largest standardized effect (+0.174 SD, $p < 0.001$), underscoring its role as a secular substitute for spiritual fulfillment. Internet activity, conversely, had a paradoxical negative association (-0.133 SD, $p < 0.001$), challenging the assumption that digital exposure amplifies success aspirations. Privilege's positive effect (+0.119 SD, $p < 0.001$) suggests that respondents who reject inherited status are slightly more success-driven, though this relationship warrants further qualitative exploration. Ultimately, the null result for religiosity implies that spiritual identity alone does not explain success attitudes in Denmark. The same R-squared score in the second and third regression show standardization does not improve explanatory power. However, it helps highlight key trends: careerism has the largest effect, followed by internet activity (negative), then privilege.

Then, the fourth regression introduces an interaction term to test whether internet usage has differential impact on religious identity. The results, visualized in Figure 1, reveal a nuanced pattern. While the slope for non-religious individuals is nearly flat, indicating that internet activity

has little impact on their success aspirations, the slope for religious individuals shows a slight decline. This suggests that religious individuals who spend more time online may place slightly less importance on success, possibly due to the distracting or disillusioning effects of digital media. However, the overlapping confidence intervals indicate that this difference is not statistically significant, and the overall effect of internet activity on success remains modest. These findings align with Moberg's (2020) hypothesis that religious individuals may be less influenced by the constant reminders of others' success online, but the lack of statistical significance calls for cautious interpretation. Future research could explore qualitative differences in how religious and non-religious individuals engage with digital media to better understand this relationship. Unfortunately, the second OLS teased a predictable outcome: the coefficient and P-value for religiosity becomes statistically insignificant, suggesting that internet usage affects attitude towards success similarly for religious and non-religious individuals.

This investigation sought to determine whether a gap exists between religious and non-religious Danish respondents in their attitudes toward success and whether that gap could be explained by differences in privilege, career aspirations, or internet exposure. Through regression analyses of ESS9 data, the findings reveal a nuanced relationship. While religious status alone proved statistically insignificant as a predictor of success attitudes ($p = 0.104$), the inclusion of covariates illuminated key drivers. This null result underscores the limitations of framing religiosity as a binary, static identity and highlights the need for more granular measures of spiritual identity in future studies. Conversely, internet usage exhibited a paradoxical negative association with success aspirations (-0.133 SD, $p < 0.001$), challenging assumptions about digital exposure

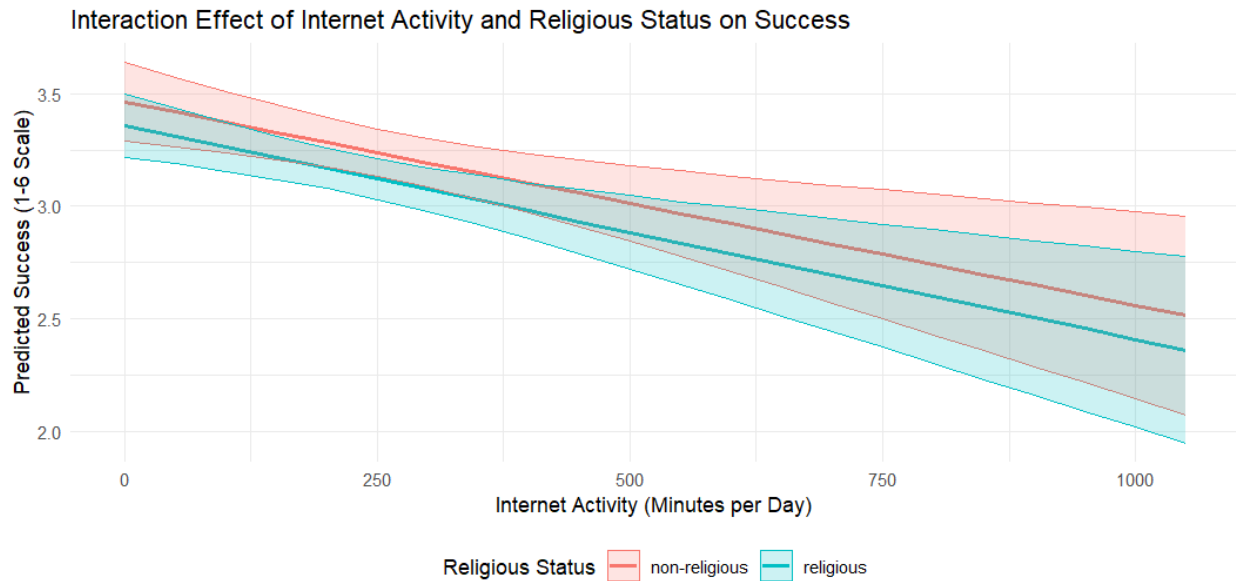
amplifying ambition. Methodologically, the observational design—while leveraging ESS9’s robust sampling—could not fully disentangle confounding variables, such as unmeasured socio-cultural influences or reverse causality. For instance, childhood autonomy or personality traits might jointly shape religiosity and success attitudes, but such data were unavailable. Ultimately, this study answers its research question in the negative: no statistically significant gap exists between religious and non-religious respondents in their valuation of success.

Table 1: Table comparing the regression with all covariables and the regression with the same variables standardized

	OLS2	ZOLS3
(Intercept)	2.580***	0.089+
	(0.189)	(0.050)
breligionreligious	−0.116	−0.086
	(0.071)	(0.053)
privilege	0.121***	
	(0.032)	
internet_activity	−0.001***	
	(0.000)	
careerism	0.348***	

	(0.079)	
pspwght	−0.055	−0.041
	(0.043)	(0.032)
z_privilege		0.119***
		(0.031)
z_internet_activit y		−0.133***
		(0.027)
z_careerism		0.174***
		(0.039)
Num.Obs.	1390	1390
R2	0.074	0.074
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		

Figure 1:



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